specialist that attacks only a few. For the specialist blight, the lethality can be turned up really high; it can knock out, say, 99 percent of a very specific group of plants. For the generalist, the range of plants attacked is much broader, but its lethality for any one plant in that range might be much smaller. That's a pattern we see again and again in Nature.

Lynda: Could you have a generalist blight that becomes much more lethal?

Meyerowitz: Something like that has happened before. Early in the Earth's history, when cyanobacteria started making oxygen, thereby changing radically the composition of the Earth's atmosphere, they managed to kill most everything else on Earth.

Leadbetter: But the oxygen was a lethal byproduct, a poison, produced by the cyanobacteria; not a generalist pathogen.

Baltimore: We may not have seen it, but I can imagine a very lethal specialist pathogen becoming a lethal generalist. It could spread the range of plants it attacks with the help of an insect that carries it to many species. A Japanese beetle, for example, which eats something like two hundred different plant species, could infect many species with the pathogen it carries, and the pathogen might adapt to attack those species, lethally.

Meyerowitz: I can conceive of a totally lethal generalist: a pathogen that attacks chloroplasts. Chloroplasts are something that all plants have in common. They are crucial to photosynthesis (the process where a plant combines sunlight with carbon dioxide from the air, and water from its roots, to produce carbohydrates that it needs for growth). Without chloroplasts, a plant will die. Now suppose that some new